Visual Memory Experiments

Experimental design Evaluation scores Detailed results

14 October 2026 | Assembly101 development data | Real video trials v5 and v6

1 Research objective

We study whether a prediction model can use past visual evidence to predict a future scene, and whether choosing what to forget can preserve useful information when memory is limited. For example, an earlier view of a toy component may remain useful after the hand or another component obscures it.

This report describes the latest real video experiments and the preceding matched trial used as a reference. It first defines the data, model, experiments and evaluation scores, then gives every reported comparison at the seed and horizon level. The interpretation is presented in the final section. Earlier synthetic Gate and Motion trials remain separate historical experiments.

2 Data and prediction model

Table 1. Fixed data used in the two real video trials

Split	Videos	Participants	Episodes	Observations	Queries
Training	58	29	136	34,816	5,168
Validation	20	10	62	15,872	2,356
Total development	78	39	198	50,688	7,524

Each episode contains 256 observations sampled 0.5 seconds apart. Prediction queries are at observation indices 40, 44, ..., 188, giving 38 queries per episode. The four prediction horizons are 1, 4, 16 and 64 observations, corresponding to 0.5, 2, 8 and 32 seconds. Training and validation participants are disjoint. The sealed test data were not evaluated.

A frozen I-JEPA encoder converts each image into a 5,120-dimensional visual feature vector. The encoder is not fine-tuned. A small causal Transformer, called the reader, is trained to predict future features from legal past observations. It uses train-only normalization and 128-dimensional input and output bases. Prediction error is measured after reconstruction in all 5,120 normalized dimensions.

Table 2. Reader and training specification

Setting	Value
Reader architecture	Hidden size 256; 2 Transformer layers; 4 attention heads; feed-forward size 1,024
Trainable parameters	1,779,072
Recent observations and historical memory	8 recent observations; historical capacity 16 for FIFO and Reservoir
Inputs	Visual features, actual observation ages, recent or history role, and prediction horizon
Independent training runs	Seeds 17, 29 and 43; 30 epochs and 9,690 optimizer steps per seed
Balanced batch	16 identical base queries repeated under all 4 conditions; 64 condition examples per step
Regularization	Model dropout 0.10; retained-history dropout 0.15
Optimizer and learning rate	AdamW; learning rate 0.0003; 5 percent warmup followed by cosine decay
Single change in v6	Weight decay increased from 0.05 to 0.20; all other scientific settings retained
Checkpoint selection	Earliest strict minimum of validation MSE averaged equally over all 4 conditions and 4 horizons

3 Experimental design

For each seed, one shared reader is trained with all four memory conditions. The comparisons change the available context while keeping that reader and the future target fixed.

Table 3. Memory conditions

Condition	Information available
Recent	The 8 most recent observations, including the current observation.
FIFO	The same recent observations plus up to 16 older observations. When historical memory fills, the oldest historical observation is deleted.
Reservoir	The same recent observations plus up to 16 older observations retained by Algorithm R. Older eligible observations can be sampled and a new eligible item can be rejected.
Full	The recent observations and all earlier available observations in the episode. No future observation is supplied.

3 1 Experiment A Future visual prediction

We evaluate all 2,356 validation queries under all 4 conditions and all 4 horizons for each of the 3 frozen readers. This gives 113,088 prediction scores. The primary comparison is Full versus Recent at 8 and 32 seconds; the 0.5 and 2 second results are also retained. Persistence predicts that the current feature remains unchanged. The fixed-basis oracle projects the known future residual into the fixed output basis and gives a diagnostic reconstruction floor; it has access to the future target.

3 2 Fixed comparison against the previous trial

The v6 Full prediction is compared with the already executed v5 Recent and v5 Full predictions for exactly the same query keys. The old models were not run again. All four horizons are reported. The predeclared main comparisons are at 8 and 32 seconds for each seed and the pooled result.

3 3 Experiment B Deletion choice and future consequences

We select 4 valid full-memory states per recording under each of 2 source trajectories, FIFO and Reservoir. All 78 development recordings are included, giving 624 states. At a state, the existing 16 older records and the next eligible record create 17 legal deletion choices. Every choice is evaluated immediately and through the same 32 subsequent Reservoir updates, with shared random draws. All 3 readers are evaluated, producing 31,824 candidate records. The reported confidence intervals use the validation participants only, covering 160 of the 624 states.

One joint oracle deletion is selected for each state and reader. Its objective is the equally weighted mean of the immediate four-horizon MSE and the mean four-horizon MSE over the 32 continuation steps. The same selected deletion is then used for the joint, 8 second and 32 second comparisons against FIFO and Reservoir deletion baselines. Exact ties delete the oldest timestamp. The oracle sees future errors, so this experiment measures available deletion headroom rather than the performance of a deployable learned selector.

3 4 Experiment C Persistent irreversible memory

All 198 episodes are replayed with FIFO, Reservoir and Uniform policies at historical capacities 8, 16 and 32. Uniform deletes one of the 17 incoming or retained candidates uniformly when capacity is 16, and uses the analogous rule at other capacities. Unlike Reservoir, Uniform gives the incoming item the same deletion chance as each currently retained item. Deleted visual payloads cannot return from an external archive. The 1,782 retention trajectories are shared by all 3 readers, giving 812,592 prediction scores at the same 38 queries and 4 horizons. Both training and validation scores are reported separately.

4 Evaluation scores and statistical interpretation

4 1 Normalized mean squared error

Let z be the 5,120-dimensional encoded feature and let mu and sigma be its training-only mean and standard deviation. Each dimension is normalized as x = (z - mu) / max(sigma, 0.000001). The reader predicts the future normalized feature x_hat. The score is the mean of (x_hat[d] - x_future[d]) squared over d = 1, ..., 5120. Lower MSE means a more accurate feature prediction. This is a visual feature error, not action classification accuracy or a percentage of correct videos.

The training loss averages this MSE equally across the four horizons and four conditions. The reader predicts output-basis coefficients; the full feature reconstruction is current normalized feature + mean future residual + predicted coefficients multiplied by the transpose of the fixed output basis.

4 2 Absolute and relative improvement

Absolute improvement Delta = baseline MSE - candidate MSE. Positive Delta means lower candidate error; negative Delta means higher candidate error. Relative improvement in percent = 100 times Delta divided by baseline MSE. The reported percentages are descriptive error reductions, not accuracy gains. For Experiment B, Delta is the reduction in its joint or horizon-specific deletion cost; no percentage is supplied because the compact summary reports differences rather than absolute baseline costs.

4 3 Joint deletion score

For deletion d and horizon h, J_h(d) = 0.5 times immediate MSE_h(d) + 0.5 times the mean of the 32 continuation MSE_h(d) values. The joint score J(d) is the mean of J_h(d) over h = 1, 4, 16 and 64. The oracle minimizes J(d) once. In the results, Delta is J(baseline) - J(oracle), or J_h(baseline) - J_h(the same oracle) for the horizon rows. A positive joint improvement is partly expected from an oracle selected on this objective; it does not show that a learned online policy can identify the choice.

4 4 Aggregation and confidence intervals

Evaluation scores first average queries within an episode, then episodes within a recording, then recordings within a participant. Participants receive equal weight. This prevents a long recording or a person with many queries from dominating the result. The validation comparisons use 10 participants. Pooled results average the three readers within each participant; they do not create 30 independent participants.

A paired participant bootstrap resamples these 10 participants 10,000 times with random seed 0. The 2.5 and 97.5 percentiles form the two-sided 95 percent confidence interval for Delta. The original intervals are reproduced directly from the executed summaries. They are not adjusted for multiple comparisons. Rows share videos and readers and must not be interpreted as independent success trials.

Table 4. Meaning of the evidence column

Code	Meaning
I	Improvement supported: the entire 95 percent interval for Delta is above zero.
U	Uncertain direction: the interval includes zero, even if the point estimate is positive.
W	Worse supported: the entire interval is below zero.

The original adoption rules required positive intervals for all primary seed and pooled comparisons in A and B, with the additional fixed old-reference checks in v6. Those rules and their historical machine outputs are retained. The overall research judgment is deferred, and the final section separates supportive evidence from uncertainty and missing semantic evaluation.

Additional policy contrasts without intervals are descriptive. The original intervals are retained, with no new statistical tests or predictions. Values are displayed to 8 decimal places, or 10 for the small deletion effects; calculations use the full stored precision.

5 Training results

Table 5. Training and validation MSE for each independently initialized reader

Seed	Best epoch	v5 best val	v6 best val	Delta best val	Gain %	v6 final train	v6 final val
17	5	0.34036500	0.33851528	+0.00184972	+0.543	0.21126342	0.38394897
29	6	0.33975386	0.33884275	+0.00091111	+0.268	0.21171161	0.38653680
43	7	0.34271761	0.34110368	+0.00161393	+0.471	0.21121550	0.38986724

Each run completed 30 epochs and 9,690 optimizer steps. Best validation MSE decreased slightly in all three runs. Selection still occurred at epochs 5, 6 and 7. The final training errors are lower than the final validation errors, and validation errors rose after the selected epochs. This is consistent with a generalization gap; it does not identify its sole cause. The selected best checkpoints, rather than the final checkpoints, were used for the scientific results below.

Training validation MSE is query-weighted and averaged across conditions and horizons. The scientific tables use participant-balanced aggregation and show separate horizons. These two kinds of means should not be directly subtracted.

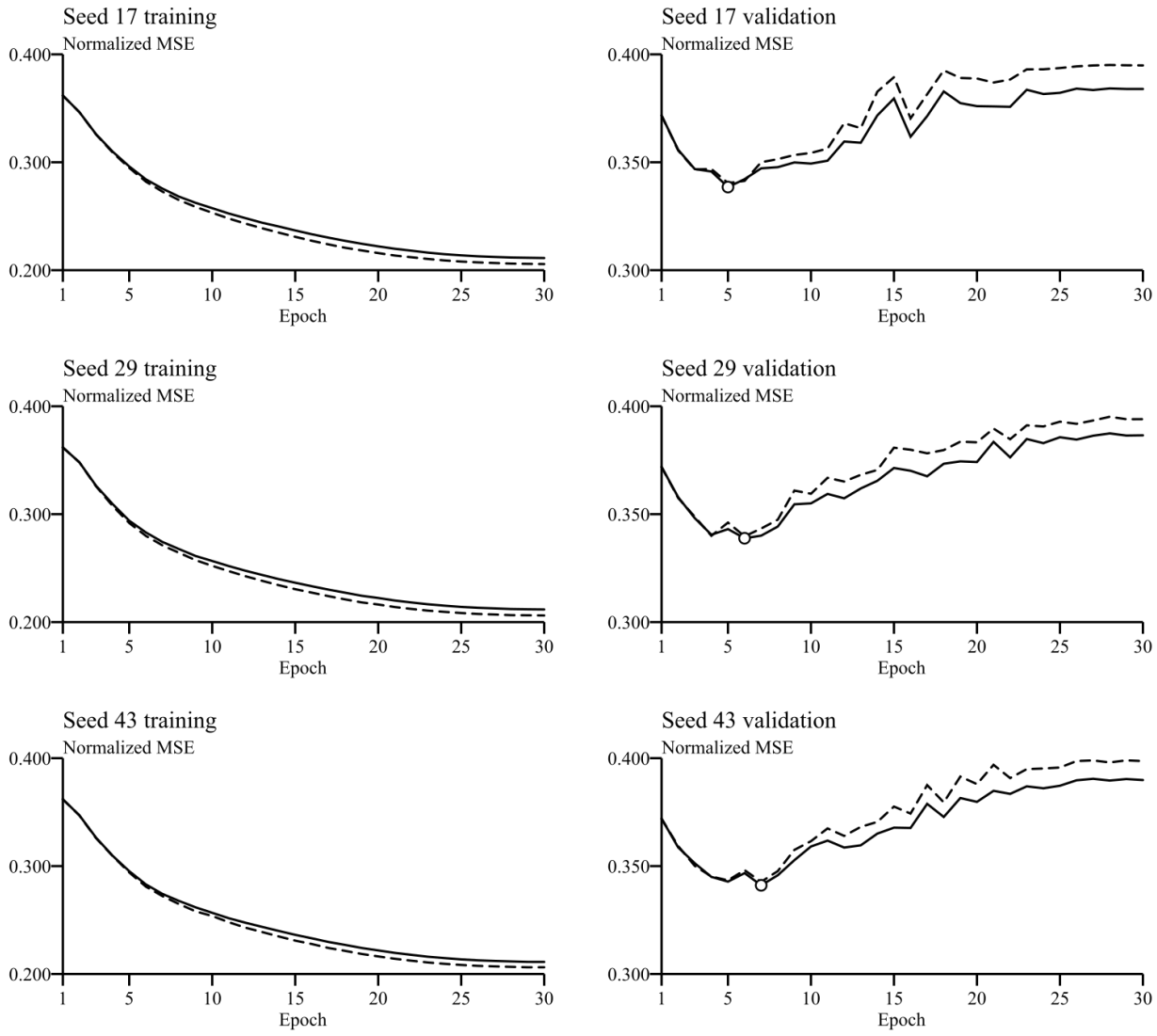


Figure 1. All 30 training and validation epochs. Dashed lines show v5 and solid lines show v6. Open circles mark the v6 selected validation minima. Exact epoch values appear in Section 10.

6 Experiment A Prediction scores

All values in this section use the validation participants and the selected best checkpoints. The Full, FIFO and Reservoir predictions share the same trained reader for a given seed.

Table 6. Absolute normalized MSE under every validation condition

Seed	Time s	Recent	FIFO	Reservoir	Full	Persistence	Basis oracle
17	0.5	0.16114383	0.16061538	0.16077397	0.16088312	0.16439188	0.05117588
29	0.5	0.16503846	0.16696935	0.16548228	0.16746271	0.16439188	0.05117588
43	0.5	0.16214346	0.16265610	0.16229208	0.16306759	0.16439188	0.05117588
pooled	0.5	0.16277525	0.16341361	0.16284945	0.16380448	0.16439188	0.05117588
17	2	0.29458287	0.29249766	0.29295104	0.29321826	0.31762146	0.07934702
29	2	0.29770624	0.29780179	0.29724047	0.29874557	0.31762146	0.07934702
43	2	0.29683444	0.29800243	0.29708982	0.29943313	0.31762146	0.07934702
pooled	2	0.29637452	0.29610063	0.29576044	0.29713232	0.31762146	0.07934702
17	8	0.40448557	0.39895852	0.40021420	0.39934402	0.44522266	0.10247927
29	8	0.39880237	0.39638468	0.39679716	0.39805944	0.44522266	0.10247927
43	8	0.40367472	0.40185643	0.40261151	0.40262738	0.44522266	0.10247927
pooled	8	0.40232089	0.39906654	0.39987429	0.40001028	0.44522266	0.10247927
17	32	0.48148398	0.47518669	0.47666551	0.47611130	0.54274519	0.12092840
29	32	0.47405732	0.47164579	0.47164376	0.47301019	0.54274519	0.12092840
43	32	0.47992963	0.47675725	0.47814285	0.47746054	0.54274519	0.12092840
pooled	32	0.47849031	0.47452991	0.47548404	0.47552734	0.54274519	0.12092840

Persistence and the basis oracle are identical across the three seeds because they use the same features and fixed fit. The oracle uses the true future feature and is only a diagnostic floor.

Table 7. Full history improvement over Recent for every seed and horizon

Seed	Time s	Recent MSE	Full MSE	Delta	Gain %	95 percent CI of Delta	Result
17	0.5	0.16114383	0.16088312	+0.00026071	+0.162	[-0.00015607, 0.00079114]	U
29	0.5	0.16503846	0.16746271	-0.00242425	-1.469	[-0.00319810, -0.00171629]	W
43	0.5	0.16214346	0.16306759	-0.00092413	-0.570	[-0.00134425, -0.00050496]	W
pooled	0.5	0.16277525	0.16380448	-0.00102923	-0.632	[-0.00145479, -0.00065011]	W
17	2.0	0.29458287	0.29321826	+0.00136462	+0.463	[-0.00013637, 0.00308674]	U
29	2.0	0.29770624	0.29874557	-0.00103933	-0.349	[-0.00198274, -0.00007947]	W
43	2.0	0.29683444	0.29943313	-0.00259868	-0.875	[-0.00381825, -0.00147138]	W
pooled	2.0	0.29637452	0.29713232	-0.00075780	-0.256	[-0.00168118, 0.00012976]	U
17	8.0	0.40448557	0.39934402	+0.00514155	+1.271	[0.00334571, 0.00751135]	I
29	8.0	0.39880237	0.39805944	+0.00074293	+0.186	[-0.00065036, 0.00187337]	U
43	8.0	0.40367472	0.40262738	+0.00104734	+0.259	[0.00038532, 0.00179583]	I
pooled	8.0	0.40232089	0.40001028	+0.00231061	+0.574	[0.00175820, 0.00298557]	I
17	32.0	0.48148398	0.47611130	+0.00537268	+1.116	[0.00308977, 0.00798169]	I
29	32.0	0.47405732	0.47301019	+0.00104713	+0.221	[-0.00040009, 0.00227796]	U
43	32.0	0.47992963	0.47746054	+0.00246908	+0.514	[0.00125971, 0.00377683]	I
pooled	32.0	0.47849031	0.47552734	+0.00296296	+0.619	[0.00200583, 0.00422645]	I

I means a positive interval, U means the interval includes zero, and W means a negative interval. In particular, the positive point estimates for seed 29 at 8 and 32 seconds retain their uncertainty.

6 1 Other memory conditions relative to Recent

This table exposes the size and direction of the FIFO and Reservoir effects at every horizon. These are arithmetic differences of the participant-balanced means. The executed Experiment A summary did not provide confidence intervals for these additional policy contrasts, so they are descriptive rather than additional statistical adoption tests.

Table 8. Every FIFO and Reservoir point improvement over Recent

Seed	Time s	Candidate	Recent MSE	Candidate MSE	Delta	Gain %
17	0.5	FIFO	0.16114383	0.16061538	+0.00052845	+0.328
17	0.5	Reservoir	0.16114383	0.16077397	+0.00036986	+0.230
29	0.5	FIFO	0.16503846	0.16696935	-0.00193089	-1.170
29	0.5	Reservoir	0.16503846	0.16548228	-0.00044382	-0.269
43	0.5	FIFO	0.16214346	0.16265610	-0.00051264	-0.316
43	0.5	Reservoir	0.16214346	0.16229208	-0.00014862	-0.092
pooled	0.5	FIFO	0.16277525	0.16341361	-0.00063836	-0.392
pooled	0.5	Reservoir	0.16277525	0.16284945	-0.00007420	-0.046
17	2.0	FIFO	0.29458287	0.29249766	+0.00208521	+0.708
17	2.0	Reservoir	0.29458287	0.29295104	+0.00163183	+0.554
29	2.0	FIFO	0.29770624	0.29780179	-0.00009556	-0.032
29	2.0	Reservoir	0.29770624	0.29724047	+0.00046577	+0.156
43	2.0	FIFO	0.29683444	0.29800243	-0.00116799	-0.393
43	2.0	Reservoir	0.29683444	0.29708982	-0.00025537	-0.086
pooled	2.0	FIFO	0.29637452	0.29610063	+0.00027389	+0.092
pooled	2.0	Reservoir	0.29637452	0.29576044	+0.00061408	+0.207
17	8.0	FIFO	0.40448557	0.39895852	+0.00552705	+1.366
17	8.0	Reservoir	0.40448557	0.40021420	+0.00427137	+1.056
29	8.0	FIFO	0.39880237	0.39638468	+0.00241768	+0.606
29	8.0	Reservoir	0.39880237	0.39679716	+0.00200521	+0.503
43	8.0	FIFO	0.40367472	0.40185643	+0.00181830	+0.450
43	8.0	Reservoir	0.40367472	0.40261151	+0.00106322	+0.263
pooled	8.0	FIFO	0.40232089	0.39906654	+0.00325434	+0.809
pooled	8.0	Reservoir	0.40232089	0.39987429	+0.00244660	+0.608
17	32.0	FIFO	0.48148398	0.47518669	+0.00629729	+1.308
17	32.0	Reservoir	0.48148398	0.47666551	+0.00481847	+1.001
29	32.0	FIFO	0.47405732	0.47164579	+0.00241153	+0.509
29	32.0	Reservoir	0.47405732	0.47164376	+0.00241355	+0.509
43	32.0	FIFO	0.47992963	0.47675725	+0.00317238	+0.661
43	32.0	Reservoir	0.47992963	0.47814285	+0.00178678	+0.372
pooled	32.0	FIFO	0.47849031	0.47452991	+0.00396040	+0.828
pooled	32.0	Reservoir	0.47849031	0.47548404	+0.00300627	+0.628

The gain is an error reduction relative to Recent. A negative number means the candidate policy had higher MSE. A larger memory condition is not assumed to be better merely because it contains older observations.

7 Comparison with the preceding real video trial

The baseline is the matching v5 reader and condition. The candidate is v6 Full. Both trials used the same fixed development data and target features. These comparisons describe a revision on an already used development set, not a new test set. The saved old scalar predictions were reused, with identical query keys and no old-model reinference.

Table 9. All matched v5 baseline and v6 Full comparisons

Seed	Times	v5 baseline	v5 MSE	v6 Full MSE	Delta	Gain %	95 percent CI of Delta	Result
17	0.5	Recent	0.16215512	0.16088312	+0.00127200	+0.784	[0.00061811, 0.00190809]	I
29	0.5	Recent	0.16397398	0.16746271	-0.00348874	-2.128	[-0.00453290, -0.00243473]	W
43	0.5	Recent	0.16338398	0.16306759	+0.00031639	+0.194	[-0.00016957, 0.00077916]	U
pooled	0.5	Recent	0.16317103	0.16380448	-0.00063345	-0.388	[-0.00103380, -0.00024670]	W
17	0.5	Full	0.16202683	0.16088312	+0.00114370	+0.706	[0.00052275, 0.00178650]	I
29	0.5	Full	0.16551658	0.16746271	-0.00194613	-1.176	[-0.00279655, -0.00110082]	W
43	0.5	Full	0.16310762	0.16306759	+0.00004003	+0.025	[-0.00140090, 0.00108985]	U
pooled	0.5	Full	0.16355034	0.16380448	-0.00025413	-0.155	[-0.00071483, 0.00019080]	U
17	2.0	Recent	0.29578177	0.29321826	+0.00256352	+0.867	[0.00152380, 0.00366681]	I
29	2.0	Recent	0.29761305	0.29874557	-0.00113251	-0.381	[-0.00349217, 0.00038375]	U
43	2.0	Recent	0.29938556	0.29943313	-0.00004756	-0.016	[-0.00209922, 0.00136681]	U
pooled	2.0	Recent	0.29759346	0.29713232	+0.00046115	+0.155	[-0.00019950, 0.00112987]	U
17	2.0	Full	0.29506014	0.29321826	+0.00184189	+0.624	[0.00030415, 0.00348030]	I
29	2.0	Full	0.29781516	0.29874557	-0.00093041	-0.312	[-0.00352276, 0.00090004]	U
43	2.0	Full	0.29993404	0.29943313	+0.00050092	+0.167	[-0.00105056, 0.00188445]	U
pooled	2.0	Full	0.29760311	0.29713232	+0.00047080	+0.158	[-0.00101845, 0.00166495]	U
17	8.0	Recent	0.40499084	0.39934402	+0.00564683	+1.394	[0.00363575, 0.00820638]	I
29	8.0	Recent	0.40037411	0.39805944	+0.00231467	+0.578	[-0.00179903, 0.00551764]	U
43	8.0	Recent	0.40646346	0.40262738	+0.00383608	+0.944	[-0.00081522, 0.00804414]	U
pooled	8.0	Recent	0.40394280	0.40001028	+0.00393252	+0.974	[0.00218606, 0.00567696]	I
17	8.0	Full	0.40144971	0.39934402	+0.00210569	+0.525	[0.00051464, 0.00400749]	I
29	8.0	Full	0.39880490	0.39805944	+0.00074547	+0.187	[-0.00300039, 0.00366544]	U
43	8.0	Full	0.40817737	0.40262738	+0.00554999	+1.360	[-0.00084758, 0.01484809]	U
pooled	8.0	Full	0.40281066	0.40001028	+0.00280038	+0.695	[0.00085118, 0.00487791]	I
17	32.0	Recent	0.48395658	0.47611130	+0.00784528	+1.621	[0.00462522, 0.01125611]	I
29	32.0	Recent	0.47721601	0.47301019	+0.00420582	+0.881	[0.00101929, 0.00748268]	I
43	32.0	Recent	0.48239016	0.47746054	+0.00492962	+1.022	[-0.00136437, 0.01108969]	U
pooled	32.0	Recent	0.48118758	0.47552734	+0.00566024	+1.176	[0.00310201, 0.00839703]	I
17	32.0	Full	0.48028756	0.47611130	+0.00417626	+0.870	[0.00163275, 0.00678611]	I
29	32.0	Full	0.47459030	0.47301019	+0.00158011	+0.333	[-0.00148899, 0.00486512]	U
43	32.0	Full	0.47963462	0.47746054	+0.00217408	+0.453	[-0.00192460, 0.00601560]	U
pooled	32.0	Full	0.47817082	0.47552734	+0.00264348	+0.553	[0.00060341, 0.00492088]	I

The original main comparisons were at 8 and 32 seconds. Shorter-horizon rows are retained here so that the reader can see favorable and unfavorable changes alike. Interpretation codes use the same paired participant interval as Experiment A.

8 Experiment B Value of selecting a deletion

Source refers to the trajectory that produced the current historical memory. Baseline refers to the deletion made at that state. All alternatives then use the same 32 Reservoir continuation steps. The joint score averages all four horizons; the 8 and 32 second scores retain the deletion selected by that joint score. The tables show every reported validation comparison, including the small uncertain effect. Delta is baseline cost minus oracle cost.

Table 10. Current source FIFO and deletion baseline FIFO

Seed	Score	Delta cost	95 percent CI of Delta	Result
17	Joint	+0.0001283443	[0.0000856677, 0.0001715346]	I
17	8 seconds	+0.0001704637	[0.0000914987, 0.0002537639]	I
17	32 seconds	+0.0002374385	[0.0001647883, 0.0003124119]	I
29	Joint	+0.0003141938	[0.0002607350, 0.0003735187]	I
29	8 seconds	+0.0003581518	[0.0002743765, 0.0004374247]	I
29	32 seconds	+0.0004102781	[0.0003305718, 0.0004918692]	I
43	Joint	+0.0001505417	[0.0000998300, 0.0002005042]	I
43	8 seconds	+0.0001827708	[0.0001082670, 0.0002637493]	I
43	32 seconds	+0.0001999007	[0.0001027997, 0.0003119096]	I
pooled	Joint	+0.0001976933	[0.0001566528, 0.0002420119]	I
pooled	8 seconds	+0.0002371288	[0.0001881095, 0.0002896165]	I
pooled	32 seconds	+0.0002825391	[0.0002158984, 0.0003545821]	I

Table 11. Current source FIFO and deletion baseline Reservoir

Seed	Score	Delta cost	95 percent CI of Delta	Result
17	Joint	+0.0001336267	[0.0000998077, 0.0001679113]	I
17	8 seconds	+0.0001800752	[0.0000858557, 0.0002746744]	I
17	32 seconds	+0.0002525876	[0.0001979696, 0.0002983540]	I
29	Joint	+0.0002571545	[0.0001970953, 0.0003169934]	I
29	8 seconds	+0.0003289653	[0.0001678219, 0.0004990270]	I
29	32 seconds	+0.0003625092	[0.0002498120, 0.0004798279]	I
43	Joint	+0.0001870026	[0.0001316756, 0.0002729015]	I
43	8 seconds	+0.0003609905	[0.0001836091, 0.0006587788]	I
43	32 seconds	+0.0001870660	[0.0000962926, 0.0002662444]	I
pooled	Joint	+0.0001925946	[0.0001541949, 0.0002339788]	I
pooled	8 seconds	+0.0002900103	[0.0001633435, 0.0004348467]	I
pooled	32 seconds	+0.0002673876	[0.0002172804, 0.0003174130]	I

8 1 Reservoir source states

Table 12. Current source Reservoir and deletion baseline FIFO

Seed	Score	Delta cost	95 percent CI of Delta	Result
17	Joint	+0.0000956823	[0.0000757205, 0.0001160763]	I
17	8 seconds	+0.0001148346	[0.0000617572, 0.0001658612]	I
17	32 seconds	+0.0001470146	[0.0001195066, 0.0001751902]	I
29	Joint	+0.0003355308	[0.0002741580, 0.0003990370]	I
29	8 seconds	+0.0002079116	[0.0000040273, 0.0003911436]	I
29	32 seconds	+0.0006665991	[0.0004889990, 0.0008764203]	I
43	Joint	+0.0001185573	[0.0000721396, 0.0001715184]	I
43	8 seconds	+0.0001262442	[0.0000714440, 0.0001798829]	I
43	32 seconds	+0.0000736861	[-0.0000007579, 0.0001566059]	U
pooled	Joint	+0.0001832568	[0.0001463190, 0.0002209172]	I
pooled	8 seconds	+0.0001496635	[0.0000650256, 0.0002270948]	I
pooled	32 seconds	+0.0002957666	[0.0002413219, 0.0003551785]	I

Table 13. Current source Reservoir and deletion baseline Reservoir

Seed	Score	Delta cost	95 percent CI of Delta	Result
17	Joint	+0.0002051544	[0.0001568429, 0.0002652676]	I
17	8 seconds	+0.0002676442	[0.0001901536, 0.0003537900]	I
17	32 seconds	+0.0003380341	[0.0002179576, 0.0005014488]	I
29	Joint	+0.0004153880	[0.0002765340, 0.0005757729]	I
29	8 seconds	+0.0005340286	[0.0002509131, 0.0008853539]	I
29	32 seconds	+0.0007944489	[0.0005278340, 0.0011019072]	I
43	Joint	+0.0001921811	[0.0001157399, 0.0002956976]	I
43	8 seconds	+0.0002509241	[0.0001681155, 0.0003397214]	I
43	32 seconds	+0.0003290580	[0.0002092982, 0.0004765362]	I
pooled	Joint	+0.0002709078	[0.0002100849, 0.0003326785]	I
pooled	8 seconds	+0.0003508657	[0.0002309755, 0.0004754093]	I
pooled	32 seconds	+0.0004871804	[0.0003735804, 0.0006100302]	I

The future-informed oracle measures headroom available in the evaluated states. Its joint advantage follows from choosing the best joint objective among all 17 candidates. Whether an online learned policy can obtain that advantage, using only available information, remains a separate research question.

9 Experiment C Continuous irreversible replay

Each table reports every seed, horizon and policy for a fixed split and capacity. FIFO MSE, Reservoir MSE and Uniform MSE are participant-balanced means. The two Delta columns subtract the alternative from FIFO; positive values favor that alternative. These are descriptive comparisons with no confidence interval supplied by the executed replay summary. They should not be described as statistically reliable improvements.

The main historical capacity is 16; capacities 8 and 32 are fixed secondary checks. The separate recent buffer contains 8 observations. Training scores use 29 participants and validation scores use 10. Pooled rows average the three readers within the same split. Training and validation scores are not directly paired with one another.

Table 14. Validation replay with historical capacity 16

Seed	Time s	FIFO MSE	Reservoir MSE	Uniform MSE	FIFO minus Reservoir	FIFO minus Uniform
17	0.5	0.16061538	0.16077397	0.16062610	-0.00015860	-0.00001072
29	0.5	0.16696935	0.16548228	0.16652128	+0.00148707	+0.00044807
43	0.5	0.16265610	0.16229208	0.16253913	+0.00036402	+0.00011697
pooled	0.5	0.16341361	0.16284945	0.16322884	+0.00056416	+0.00018477
17	2.0	0.29249766	0.29295104	0.29253513	-0.00045338	-0.00003747
29	2.0	0.29780179	0.29724047	0.29758799	+0.00056133	+0.00021380
43	2.0	0.29800243	0.29708982	0.29769956	+0.00091261	+0.00030287
pooled	2.0	0.29610063	0.29576044	0.29594089	+0.00034019	+0.00015973
17	8.0	0.39895852	0.40021420	0.39913818	-0.00125567	-0.00017965
29	8.0	0.39638468	0.39679716	0.39628232	-0.00041247	+0.00010236
43	8.0	0.40185643	0.40261151	0.40188984	-0.00075508	-0.00003342
pooled	8.0	0.39906654	0.39987429	0.39910345	-0.00080774	-0.00003690
17	32.0	0.47518669	0.47666551	0.47537732	-0.00147882	-0.00019063
29	32.0	0.47164579	0.47164376	0.47141331	+0.00000203	+0.00023248
43	32.0	0.47675725	0.47814285	0.47689101	-0.00138561	-0.00013377
pooled	32.0	0.47452991	0.47548404	0.47456055	-0.00095413	-0.00003064

Table 15. Validation replay with historical capacity 8

Seed	Time s	FIFO MSE	Reservoir MSE	Uniform MSE	FIFO minus Reservoir	FIFO minus Uniform
17	0.5	0.16061992	0.16080359	0.16062943	-0.00018367	-0.00000952
29	0.5	0.16640996	0.16522443	0.16605624	+0.00118553	+0.00035372
43	0.5	0.16244667	0.16220268	0.16236018	+0.00024400	+0.00008650
pooled	0.5	0.16315885	0.16274356	0.16301528	+0.00041529	+0.00014357
17	2.0	0.29267041	0.29323044	0.29269206	-0.00056004	-0.00002165
29	2.0	0.29761146	0.29717446	0.29736635	+0.00043700	+0.00024511
43	2.0	0.29742913	0.29692023	0.29726829	+0.00050890	+0.00016084
pooled	2.0	0.29590366	0.29577504	0.29577557	+0.00012862	+0.00012810
17	8.0	0.39969107	0.40117660	0.39975713	-0.00148553	-0.00006607
29	8.0	0.39631312	0.39720741	0.39628703	-0.00089429	+0.00002609
43	8.0	0.40195638	0.40295856	0.40199401	-0.00100218	-0.00003762
pooled	8.0	0.39932019	0.40044752	0.39934606	-0.00112733	-0.00002587
17	32.0	0.47605749	0.47781577	0.47614900	-0.00175827	-0.00009151
29	32.0	0.47143370	0.47226616	0.47127618	-0.00083245	+0.00015752
43	32.0	0.47704152	0.47877879	0.47718085	-0.00173727	-0.00013932
pooled	32.0	0.47484424	0.47628690	0.47486868	-0.00144266	-0.00002444

9 Experiment C Continued replay scores

Table 16. Validation replay with historical capacity 32

Seed	Time s	FIFO MSE	Reservoir MSE	Uniform MSE	FIFO minus Reservoir	FIFO minus Uniform
17	0.5	0.16069347	0.16075017	0.16068410	-0.00005670	+0.00000937
29	0.5	0.16729537	0.16598621	0.16689182	+0.00130916	+0.00040355
43	0.5	0.16286933	0.16249896	0.16273686	+0.00037037	+0.00013247
pooled	0.5	0.16361939	0.16307844	0.16343759	+0.00054094	+0.00018179
17	2.0	0.29263669	0.29275149	0.29258313	-0.00011480	+0.00005356
29	2.0	0.29803160	0.29743519	0.29778969	+0.00059641	+0.00024191
43	2.0	0.29869098	0.29756499	0.29827691	+0.00112600	+0.00041407
pooled	2.0	0.29645309	0.29591722	0.29621658	+0.00053587	+0.00023651
17	8.0	0.39879488	0.39945655	0.39885624	-0.00066167	-0.00006136
29	8.0	0.39673931	0.39660011	0.39652801	+0.00013920	+0.00021129
43	8.0	0.40210803	0.40236909	0.40204491	-0.00026106	+0.00006312
pooled	8.0	0.39921407	0.39947525	0.39914306	-0.00026118	+0.00007102
17	32.0	0.47493818	0.47575811	0.47500601	-0.00081993	-0.00006783
29	32.0	0.47222754	0.47155415	0.47181477	+0.00067340	+0.00041277
43	32.0	0.47692990	0.47755163	0.47690196	-0.00062174	+0.00002793
pooled	32.0	0.47469854	0.47495463	0.47457425	-0.00025609	+0.00012429

Table 17. Train replay with historical capacity 16

Seed	Time s	FIFO MSE	Reservoir MSE	Uniform MSE	FIFO minus Reservoir	FIFO minus Uniform
17	0.5	0.15618821	0.15640200	0.15622467	-0.00021379	-0.00003646
29	0.5	0.15378208	0.15337419	0.15366058	+0.00040789	+0.00012150
43	0.5	0.15027854	0.15018352	0.15024812	+0.00009503	+0.00003042
pooled	0.5	0.15341628	0.15331990	0.15337779	+0.00009638	+0.00003849
17	2.0	0.25808282	0.25937149	0.25834331	-0.00128867	-0.00026049
29	2.0	0.24703091	0.24723118	0.24696688	-0.00020026	+0.00006403
43	2.0	0.23836923	0.23874462	0.23844021	-0.00037539	-0.00007098
pooled	2.0	0.24782766	0.24844910	0.24791680	-0.00062144	-0.00008915
17	8.0	0.32372068	0.32409505	0.32376240	-0.00037437	-0.00004172
29	8.0	0.31078355	0.31001550	0.31046575	+0.00076805	+0.00031780
43	8.0	0.29765497	0.29767528	0.29763674	-0.00002031	+0.00001823
pooled	8.0	0.31071973	0.31059528	0.31062163	+0.00012446	+0.00009810
17	32.0	0.38742552	0.38780028	0.38747159	-0.00037476	-0.00004608
29	32.0	0.37048399	0.36965747	0.37010827	+0.00082652	+0.00037572
43	32.0	0.35426384	0.35426210	0.35424424	+0.00000174	+0.00001960
pooled	32.0	0.37072445	0.37057328	0.37060803	+0.00015117	+0.00011641

9 Experiment C Continued replay scores

Table 18. Train replay with historical capacity 8

Seed	Time s	FIFO MSE	Reservoir MSE	Uniform MSE	FIFO minus Reservoir	FIFO minus Uniform
17	0.5	0.15630843	0.15657409	0.15634500	-0.00026565	-0.00003656
29	0.5	0.15364827	0.15337492	0.15355056	+0.00027335	+0.00009771
43	0.5	0.15023091	0.15018929	0.15021090	+0.00004162	+0.00002001
pooled	0.5	0.15339587	0.15337943	0.15336882	+0.00001644	+0.00002705
17	2.0	0.25883614	0.26002992	0.25899860	-0.00119378	-0.00016246
29	2.0	0.24700974	0.24741634	0.24697731	-0.00040660	+0.00003243
43	2.0	0.23845665	0.23884154	0.23852008	-0.00038490	-0.00006343
pooled	2.0	0.24810084	0.24876260	0.24816533	-0.00066176	-0.00006449
17	8.0	0.32384478	0.32442355	0.32386145	-0.00057877	-0.00001667
29	8.0	0.31044435	0.31006677	0.31023243	+0.00037758	+0.00021192
43	8.0	0.29761678	0.29771639	0.29761232	-0.00009962	+0.00000446
pooled	8.0	0.31063530	0.31073557	0.31056873	-0.00010027	+0.00006657
17	32.0	0.38755214	0.38818732	0.38758760	-0.00063519	-0.00003546
29	32.0	0.37015870	0.36971628	0.36989231	+0.00044242	+0.00026638
43	32.0	0.35426160	0.35432730	0.35422213	-0.00006570	+0.00003947
pooled	32.0	0.37065748	0.37074364	0.37056735	-0.00008616	+0.00009013

Table 19. Train replay with historical capacity 32

Seed	Time s	FIFO MSE	Reservoir MSE	Uniform MSE	FIFO minus Reservoir	FIFO minus Uniform
17	0.5	0.15615185	0.15627202	0.15618187	-0.00012017	-0.00003001
29	0.5	0.15386595	0.15346905	0.15374769	+0.00039690	+0.00011826
43	0.5	0.15033919	0.15020804	0.15029470	+0.00013115	+0.00004449
pooled	0.5	0.15345233	0.15331637	0.15340809	+0.00013596	+0.00004425
17	2.0	0.25754440	0.25859667	0.25782262	-0.00105227	-0.00027822
29	2.0	0.24714703	0.24709830	0.24704002	+0.00004872	+0.00010701
43	2.0	0.23841207	0.23865074	0.23843111	-0.00023867	-0.00001904
pooled	2.0	0.24770116	0.24811524	0.24776458	-0.00041407	-0.00006342
17	8.0	0.32398391	0.32399919	0.32391701	-0.00001528	+0.00006690
29	8.0	0.31104049	0.31007498	0.31071684	+0.00096551	+0.00032365
43	8.0	0.29773314	0.29767634	0.29767786	+0.00005680	+0.00005528
pooled	8.0	0.31091918	0.31058350	0.31077057	+0.00033567	+0.00014861
17	32.0	0.38761918	0.38764620	0.38755274	-0.00002702	+0.00006644
29	32.0	0.37076587	0.36985328	0.37038516	+0.00091260	+0.00038072
43	32.0	0.35432055	0.35423049	0.35427013	+0.00009006	+0.00005042
pooled	32.0	0.37090187	0.37057666	0.37073601	+0.00032521	+0.00016586

The execution audit checked all 456,192 trace rows: fixed random draws, capacity, chronological ages and no reentry of deleted payloads. This establishes faithful execution of the replay. It is distinct from an effect size or a test of action understanding.

9 1 Evaluation that remains unavailable

The raw RGB videos are at 60 frames per second, while the fine action annotations refer to a 30 frame per second sequence. The original producer and its exact starting frame have not been established. A selected raw video contains 21,277 frames, whereas the associated capture timestamp metadata contain 21,276 entries and no embedded capture timecode resolves the annotation origin. A consumer that reads 30 fps labels is not proof of the raw frame used for annotation frame zero.

Table 20. Alignment sensitivity under two unverified sampling phases

Split	Different verb sets	Observations	Fraction percent
Training	1,234	34,816	3.544
Validation	493	15,872	3.106
All development	1,727	50,688	3.407

This sensitivity calculation does not establish which phase is correct. Therefore no exact label-to-feature join was selected. Verb prediction and negative log likelihood, within-activity drift, response to an activity switch, different-activity donor corruption, and recovery at 0, 1, 2, 4, 8 and 16 steps remain unavailable. The visual MSE experiments above do not require action labels and their numerical results remain valid within that scope.

9 2 Execution and evidence records

The v6 training job completed successfully on one NVIDIA A100 with four scientific threads in 50 minutes 14 seconds. The fixed evaluation completed successfully in 28 minutes 10 seconds using four scientific CPU threads. Before inference, all six best and latest checkpoints were checked for completion, finite weights, source and cache binding, optimizer settings, schedule, random state and the earliest selected validation minimum. The final audit repeated the saved proof and verified all 78 feature files and the reported output hashes.

Independent numerical replay of the stored raw scores reproduced the existing summaries and intervals. This was a numerical audit of the same predictions, not another experiment or an independent replication. Raw model weights, features and large scalar or trace files remain on the course HPC. No additional science was run to produce this report.

Table 21. Traceable compact source records

Record	SHA256
Training report	e7da2f91dbc05f9cd5c6ab349c63f60848ca79da8059674db378bc6c62f50b73
Diagnostic protocol	07594ef120ac2e32c0e56d22701f8365fda44447089b86bde5cb55e157455222
Visual health summary	380c887ff32e2621bac2114477af838e0eebf93f9ef728529eb45bb2450dc89b
Fixed previous trial comparison	805cd6776b5997dcbc41aa002bfb166ccf6aca4ea627c02b26a62860f285bbdf
Deletion summary	241e47199a153b625855065ae86c0e27deed207e112aab2b864e37aa9fd6a9ed
Replay participant summary	0f36627f4f8618eb766ec63018100a3877512519617cd287d2465e7712fca85c
Final source and numeric audit	8ff4208ba1ac66d21a36543d7c2c525d5ac80885269ddaa7e48ee36f47a7f7a8

Source folder: runs/phase1_v6_revision/native7251_7268. Previous-trial means and training curves are in runs/phase1_v5_real_evaluation/native_completed7209_diagnostic7226. The epoch tables that follow retain every training and validation mean before the final interpretation.

10 Exact training trajectory for seed 17

Each row is one completed epoch. Train and validation columns are the full normalized MSE averaged across the four conditions and horizons. Delta is v5 minus v6 at the same epoch; positive values favor v6. These point differences have no confidence interval. An asterisk marks the selected validation minimum within that trial.

Table 22. All 30 epochs for seed 17

Epoch	v5 train	v6 train	Delta train	v5 validation	v6 validation	Delta validation
1	0.36183374	0.36183091	+0.00000283	0.37166732	0.37166733	-0.00000001
2	0.34632317	0.34640969	-0.00008652	0.35545670	0.35577840	-0.00032170
3	0.32575824	0.32599693	-0.00023869	0.34668320	0.34684250	-0.00015930
4	0.30899320	0.30986917	-0.00087597	0.34688466	0.34572148	+0.00116318
5	0.29478011	0.29612098	-0.00134087	0.34036500*	0.33851528*	+0.00184972
6	0.28207750	0.28420944	-0.00213194	0.34136929	0.34218627	-0.00081699
7	0.27277733	0.27558077	-0.00280344	0.35001373	0.34718076	+0.00283297
8	0.26493518	0.26813862	-0.00320344	0.35145267	0.34772489	+0.00372778
9	0.25862747	0.26239506	-0.00376759	0.35337278	0.34991869	+0.00345409
10	0.25321279	0.25745986	-0.00424707	0.35431788	0.34938426	+0.00493362
11	0.24783908	0.25250217	-0.00466308	0.35635039	0.35072079	+0.00562959
12	0.24315802	0.24821598	-0.00505796	0.36809995	0.35959899	+0.00850095
13	0.23887260	0.24400078	-0.00512818	0.36579659	0.35904303	+0.00675355
14	0.23482510	0.24052095	-0.00569585	0.38272415	0.37172376	+0.01100039
15	0.23092673	0.23681820	-0.00589147	0.38942206	0.37953655	+0.00988552
16	0.22723331	0.23330026	-0.00606695	0.37033714	0.36184276	+0.00849439
17	0.22391011	0.23014905	-0.00623893	0.38151120	0.37139012	+0.01012108
18	0.22077437	0.22719417	-0.00641981	0.39258938	0.38280827	+0.00978111
19	0.21831769	0.22450005	-0.00618235	0.38905974	0.37739179	+0.01166795
20	0.21586655	0.22208267	-0.00621611	0.38880039	0.37599350	+0.01280689
21	0.21356939	0.21982713	-0.00625775	0.38693840	0.37587006	+0.01106834
22	0.21198703	0.21807167	-0.00608464	0.38835068	0.37569157	+0.01265912
23	0.21040597	0.21625723	-0.00585127	0.39304635	0.38359671	+0.00944964
24	0.20901931	0.21481462	-0.00579531	0.39306958	0.38162243	+0.01144715
25	0.20797953	0.21375009	-0.00577057	0.39365676	0.38217256	+0.01148421
26	0.20723195	0.21287332	-0.00564137	0.39442217	0.38411891	+0.01030326
27	0.20665085	0.21227367	-0.00562283	0.39485004	0.38346231	+0.01138773
28	0.20610743	0.21174551	-0.00563807	0.39506221	0.38420761	+0.01085460
29	0.20586146	0.21149907	-0.00563761	0.39492786	0.38396355	+0.01096431
30	0.20566923	0.21126342	-0.00559419	0.39486956	0.38394897	+0.01092060

The selected checkpoint uses the original earliest validation minimum rule. The same-epoch differences in this table describe optimization and generalization behavior; they do not replace the paired participant evaluation of selected checkpoints.

10 Exact training trajectory for seed 29

Each row is one completed epoch. Train and validation columns are the full normalized MSE averaged across the four conditions and horizons. Delta is v5 minus v6 at the same epoch; positive values favor v6. These point differences have no confidence interval. An asterisk marks the selected validation minimum within that trial.

Table 23. All 30 epochs for seed 29

Epoch	v5 train	v6 train	Delta train	v5 validation	v6 validation	Delta validation
1	0.36184696	0.36184409	+0.00000287	0.37189464	0.37189872	-0.00000408
2	0.34796887	0.34801140	-0.00004252	0.35747258	0.35786435	-0.00039177
3	0.32582796	0.32634628	-0.00051832	0.34879152	0.34814732	+0.00064419
4	0.30757601	0.30937179	-0.00179578	0.33999733	0.34047580	-0.00047847
5	0.29181356	0.29394464	-0.00213108	0.34610720	0.34308948	+0.00301771
6	0.27991689	0.28311684	-0.00319995	0.33975386*	0.33884275*	+0.00091111
7	0.27112289	0.27426278	-0.00313989	0.34333896	0.34004697	+0.00329200
8	0.26419160	0.26765925	-0.00346765	0.34744549	0.34425738	+0.00318811
9	0.25738307	0.26124036	-0.00385729	0.36093847	0.35458014	+0.00635833
10	0.25209820	0.25661702	-0.00451882	0.35946270	0.35501037	+0.00445233
11	0.24720264	0.25190039	-0.00469775	0.36684303	0.35938540	+0.00745762
12	0.24252204	0.24758652	-0.00506448	0.36512076	0.35734657	+0.00777419
13	0.23826674	0.24370324	-0.00543650	0.36824207	0.36191877	+0.00632330
14	0.23422847	0.23985525	-0.00562678	0.37048538	0.36547721	+0.00500817
15	0.23048617	0.23646602	-0.00597986	0.38079318	0.37140142	+0.00939176
16	0.22730113	0.23321685	-0.00591572	0.37985305	0.37010497	+0.00974808
17	0.22405032	0.23010104	-0.00605072	0.37819818	0.36756207	+0.01063611
18	0.22105009	0.22726151	-0.00621142	0.37969930	0.37336042	+0.00633889
19	0.21825131	0.22440100	-0.00614969	0.38360767	0.37447121	+0.00913646
20	0.21629796	0.22234437	-0.00604641	0.38329386	0.37415549	+0.00913837
21	0.21395093	0.21997557	-0.00602464	0.38965234	0.38351392	+0.00613842
22	0.21218944	0.21818377	-0.00599433	0.38473622	0.37631745	+0.00841877
23	0.21061637	0.21651100	-0.00589463	0.39115702	0.38482728	+0.00632973
24	0.20947537	0.21521521	-0.00573984	0.39063426	0.38291006	+0.00772419
25	0.20840980	0.21406475	-0.00565495	0.39285464	0.38565164	+0.00720301
26	0.20754592	0.21326998	-0.00572406	0.39185938	0.38455674	+0.00730264
27	0.20704390	0.21270452	-0.00566062	0.39340096	0.38634733	+0.00705363
28	0.20643407	0.21210121	-0.00566714	0.39513623	0.38742355	+0.00771268
29	0.20631614	0.21187454	-0.00555841	0.39400342	0.38642977	+0.00757365
30	0.20612807	0.21171161	-0.00558353	0.39401980	0.38653680	+0.00748300

The selected checkpoint uses the original earliest validation minimum rule. The same-epoch differences in this table describe optimization and generalization behavior; they do not replace the paired participant evaluation of selected checkpoints.

10 Exact training trajectory for seed 43

Each row is one completed epoch. Train and validation columns are the full normalized MSE averaged across the four conditions and horizons. Delta is v5 minus v6 at the same epoch; positive values favor v6. These point differences have no confidence interval. An asterisk marks the selected validation minimum within that trial.

Table 24. All 30 epochs for seed 43

Epoch	v5 train	v6 train	Delta train	v5 validation	v6 validation	Delta validation
1	0.36190808	0.36190559	+0.00000249	0.37195002	0.37195675	-0.00000673
2	0.34684006	0.34698629	-0.00014624	0.35927243	0.35865013	+0.00062230
3	0.32587897	0.32633465	-0.00045567	0.35008654	0.35130966	-0.00122313
4	0.30927908	0.30957324	-0.00029417	0.34517547	0.34499049	+0.00018498
5	0.29411444	0.29527726	-0.00116282	0.34339705	0.34280466	+0.00059239
6	0.28097954	0.28305610	-0.00207656	0.34811440	0.34674504	+0.00136936
7	0.27217083	0.27420847	-0.00203764	0.34271761*	0.34110368*	+0.00161393
8	0.26477998	0.26768864	-0.00290866	0.34766702	0.34575135	+0.00191566
9	0.25804712	0.26177548	-0.00372836	0.35750050	0.35273838	+0.00476212
10	0.25374679	0.25668479	-0.00293800	0.36154548	0.35907887	+0.00246661
11	0.24796468	0.25167251	-0.00370783	0.36745344	0.36180940	+0.00564404
12	0.24294376	0.24751979	-0.00457603	0.36398615	0.35858458	+0.00540157
13	0.23892584	0.24370253	-0.00477669	0.36816540	0.35962364	+0.00854176
14	0.23483474	0.23993998	-0.00510524	0.37047461	0.36505739	+0.00541723
15	0.23093070	0.23628928	-0.00535858	0.37757390	0.36780270	+0.00977120
16	0.22779112	0.23307492	-0.00528379	0.37433506	0.36766377	+0.00667129
17	0.22413317	0.22969198	-0.00555881	0.38752100	0.37883976	+0.00868123
18	0.22144818	0.22698634	-0.00553816	0.37951474	0.37275409	+0.00676065
19	0.21859624	0.22413401	-0.00553777	0.39148747	0.38152982	+0.00995765
20	0.21629047	0.22184581	-0.00555534	0.38796719	0.37975522	+0.00821198
21	0.21411354	0.21956864	-0.00545511	0.39691678	0.38486777	+0.01204901
22	0.21233097	0.21776508	-0.00543411	0.39077499	0.38345359	+0.00732140
23	0.21067580	0.21599925	-0.00532346	0.39493636	0.38694834	+0.00798803
24	0.20942423	0.21474513	-0.00532091	0.39521738	0.38604422	+0.00917316
25	0.20839680	0.21354747	-0.00515067	0.39566132	0.38723037	+0.00843095
26	0.20754843	0.21269183	-0.00514340	0.39866552	0.38975058	+0.00891494
27	0.20706613	0.21212266	-0.00505653	0.39896616	0.39044888	+0.00851728
28	0.20663561	0.21166296	-0.00502735	0.39800416	0.38963016	+0.00837400
29	0.20621473	0.21124675	-0.00503202	0.39899039	0.39033591	+0.00865448
30	0.20622264	0.21121550	-0.00499286	0.39865064	0.38986724	+0.00878339

The selected checkpoint uses the original earliest validation minimum rule. The same-epoch differences in this table describe optimization and generalization behavior; they do not replace the paired participant evaluation of selected checkpoints.

11 Summary and interpretation

The experiments provide measurable evidence that historical visual context can help longer-horizon feature prediction on the fixed validation recordings. For the pooled v6 readers, the 8 second MSE decreases from 0.40232089 to 0.40001028, an absolute decrease of 0.00231061 and a relative decrease of +0.574 percent. Its 95 percent interval is [0.00175820, 0.00298557]. At 32 seconds the pooled error decreases from 0.47849031 to 0.47552734, with relative decrease +0.619 percent and interval [0.00200583, 0.00422645].

Training variation still matters. Seeds 17 and 43 have positive history-benefit intervals at both longer horizons; seed 29 has small positive estimates with intervals crossing zero. At 0.5 and 2 seconds, Full has no supported positive advantage over Recent. The pooled 0.5 second difference is negative, and the pooled 2 second interval crosses zero. Historical context therefore has a horizon-dependent effect in this experiment.

The deletion experiment shows that many evaluated memory states have room for a better deletion choice. For example, under a Reservoir source and Reservoir baseline, the pooled oracle decreases joint cost by 0.0002709078, with interval [0.0002100849, 0.0003326785]. The smallest uncertain comparison is Reservoir source, FIFO baseline, seed 43 at 32 seconds: Delta 0.0000736861, interval [-0.0000007579, 0.0001566059]. These effects concern a future-informed oracle and do not yet establish that a learned online selector can obtain them.

Increasing weight decay from 0.05 to 0.20 slightly improves the best validation mean in every training run, but the matched previous-trial tables show that the change is not uniformly better across seeds and horizons. Replay is complete and respects irreversible deletion; its policy scores are descriptive and often close. A faithful replay audit should not be interpreted as a separate performance success.

As a compact count of the detailed evidence, 6 of the 8 original longer-horizon history comparisons have positive intervals; 9 of the 16 original longer-horizon previous-trial comparisons do; and 47 of the 48 deletion comparisons do. These rows share readers and participants. The counts are not an empirical probability of success and are not a substitute for the effect sizes, denominators and intervals reported above.

The present interpretation is partial support for longer-horizon visual memory and for oracle deletion headroom, with incomplete robustness across training runs and unresolved action-semantic evaluation. We defer an overall pass or fail judgment for the research. The original numerical outputs and adoption rules remain preserved. Completion of the visual calculations does not establish action understanding, a deployable learned deletion policy, or independent held-out confirmation.